CH 3 — Series

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Info — Infinite Series

An infinite series is an expression of the form

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + ...$$

where $\{a_n\}$ is a sequence.

- the sequence of terms $\{a_n\}_{n=1}^{\infty}$
- the sequence of partial sums $\{S_m\}_{m=1}^{\infty}$ where S_m is the truncated sum up to the term a_m:

$$S_m = a_1 + a_2 + a_3 + ... + a_m$$

We say that a series converges if the sequence of partial sums converges.

If $\lim_{m\to\infty} S_m = S$, then S is called the sum of the series.

Otherwise the series diverges.

Note that oscillating result is assigned with "no limit" or DNE

Geometric Series

Info — Harmonic Series

The harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

is always divergent.

Info — Geometric Series

Let $a, r \in \mathbb{R}$ s.t. $a \neq 0$. A geometric series is a series in form

$$\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + ...$$

a is called the first term of the geometric series and r is the ratio of the geometric series.

The partial sum

$$S_m = \sum_{n=0}^{m} ar^n = a + ar + ar^2 + ... + ar^m = a\left(\frac{1 - r^{m+1}}{1 - r}\right)$$

where $r \neq 1$

Info — Geometric Series Test

Let $\sum_{n=0}^{\infty} ar^n$ be a geometric series, where $a \neq 0$

1. If $|r| < 1$, then $\sum_{n=0}^{\infty} ar^n$ converges and $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$
2. If $|r| \geq 1$, then $\sum_{n=0}^{\infty} ar^n$ diverges

Examples:

1. $\sum_{n=1}^{\infty} \frac{1}{n}$
   - Prove that the partial sums of the series satisfy $S_{2^k} \geq 1 + \frac{k}{2}$ for all non-ngative integers k

     $k = 0 \Longrightarrow S_{2^0} = 1 = 1 + \frac{0}{2}$

     $k = 1 \Longrightarrow S_2 = 1 + \frac{1}{2} = 1 + \frac{1}{2}$

     $k = 2 \Longrightarrow S_4 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} > S_2 + \frac{1}{2} = 2 = 2$

     $k = 3 \Longrightarrow S_8 = S_4 + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} > S_4 + \frac{1}{2} = 1 + \frac{3}{2} = 1 + \frac{3}{2}$
2. $\sum_{n=0}^{\infty} \frac{1}{2}\left(\frac{1}{2}\right)^n$ converges to $\frac{\frac{1}{2}}{\frac{1}{2}} = 1$ by GST
3. $\sum_{n=0}^{\infty} \left(-\frac{4}{3}\right)^n$ diverges by GST

Info — Arithemtic Properties

Assume that the series $\sum_{n=1}^{\infty} a_n, \sum_{n=1}^{\infty} b_n$ both converges

1. $\sum_{n=1}^{\infty} ca_n$ converges for every $c \in \mathbb{R}$, $\sum_{n=1}^{\infty} ca_n = c\sum_{n=1}^{\infty} a_n$
2. $\sum_{n=1}^{\infty} a_n + b_n$ converges, $\sum_{n=1}^{\infty} a_n + b_n = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$

Info — Tail Convergence

Given an infinite series $\sum_{n=1}^{\infty} a_n$, let $S_m = \sum_{n=1}^{m} a_n$ denote its m-th partial sum. The difference

$$\sum_{n=1}^{\infty} a_n - S_m = a_{m+1} + a_{m+2} + a_{m+3} + \ldots = \sum_{n=m+1}^{\infty} a_n$$

is caled a tail of the infinite series.

Let $\{a_n\}_{n=1}^{\infty}$ be a sequnce and let j denote a positive integer.

1. If $\sum_{n=1}^{\infty} a_n$ converges, then its tail also converges for each $j \geq 1$
2. If the tail $\sum_{n=1}^{\infty} a_n$ converges for some j, then $\sum_{n=1}^{\infty} a_n$ converges.

Warning — Necessary Condition for Convergence

If $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

The converse is false, the contrapositive is called the **Divergence Test**

> **Info — Divergence Test**
>
> If $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Examples:

1. $\sum_{n=0}^{\infty} \frac{(-3)^{2n}}{2^{3n-3}} = \sum_{n=0}^{\infty} 8\left(\frac{9}{8}\right)^n$, since$\left(\left|\frac{9}{8}\right| > 1\right)$, divergent by GST
2. $\sum_{n=1}^{\infty} (-2)^n 5^{2-n} = \sum_{n=1}^{\infty} 25\left(-\frac{2}{5}\right)^n = \sum_{n=0}^{\infty} -10\left(-\frac{2}{5}\right)^n$, since $\left|-\frac{2}{5}\right| < 1$, converging to $-\frac{10}{1+\frac{2}{5}} = -\frac{50}{7}$

Telescoping Series

Example:

$\sum_{n=1}^{\infty} \frac{1}{n} - \frac{1}{n+1}$

$S_1 = a_1 = 1 - \frac{1}{2}$

$S_2 = a_1 + a_2 = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) = 1 - \frac{1}{3}$

By induction, we can prove that $\forall m \in \mathbb{N}, S_m = 1 - \frac{1}{m+1}$

Convergence Tests for series with Nonnegative Terms

> **Warning — Positive Series**
>
> We call a series $\sum_{n=1}^{\infty} a_n$ positive if $a_n \geq 0, \forall n \geq 1$
>
> If there is some positive integer j s.t. $a_n \geq 0, \forall n \geq j$, then the series is said to be eventually positive. The sequence of partial sum is a montone sequence that is non-decreasing.
>
> 1. If the sequence is bounded above, the series converges
> 2. If the sequence is not bounded above, the series diverges to ∞

> **Info — Integral Test**
>
> Let f be a function that is defined on the infinite interval $[1, \infty)$. Suppose that f is continuous, positive, and **decreasing** on the subinterval $[k, \infty)$ for some positive integer k. Then the following statements hold.
>
> 1. If $\int_k^{\infty} f(x)\,\mathrm{d}x$ converges, then the series $\sum_{n=1}^{\infty} f(n)$ converges
> 2. If $\int_k^{\infty} f(x)\,\mathrm{d}x$ diverges, then the series $\sum_{n=1}^{\infty} f(n)$ diverges to ∞

Examples:

1. $\sum_{n=1}^{\infty} \frac{1}{n}$. Let $f(x) = \frac{1}{n}$. We have $\int_1^{\infty} \frac{1}{x}\,\mathrm{d}x$ have the same behaviour as of the series. $\lim_{t\to\infty} \int_1^t \frac{1}{x}\,\mathrm{d}x = \lim_{t\to\infty} \ln x\Big|_1^{\infty} = \infty$. Since the limit of the integral diverges, the series diverges to ∞ by the Integral Test
2. $\sum_{n=1}^{\infty} \frac{1}{n^2}$. Let $f(x) = \frac{1}{n^2}$. We have $\int_1^{\infty} \frac{1}{x^2}\,\mathrm{d}x$ have the same behaviour as of the series. $\lim_{t\to\infty} \int_1^t \frac{1}{x^2}\,\mathrm{d}x = \lim_{t\to\infty} -\frac{1}{x}\Big|_1^{\infty} = 0 - (-1) = 1$. Since the limit of the integral converges, the series converges by the Integral Test

📐 **Theorem — Integral Test Estimation Theorem**

Suppose that a function f is continuous, positive, and decreasing on the infinite interval $[k, \infty)$ for some positive integer k. Further suppose that $\int_k^\infty f(x)\,dx$ converges so that $\sum_{n=1}^\infty f(n) = S$ for some real number S by the Integral Test. Let $S_m = \sum_{n=1}^m f(n)$ denote the m —th partial sum. Then for any integer $m \geq k$, we have

$$\int_{m+1}^\infty f(x)\,dx \leq S - S_m \leq \int_m^\infty f(x)\,dx$$

Example:

$\sum_{i=1}^\infty \frac{\ln n}{n}$.

Let $f(x) = \frac{\ln x}{x}$

f is continuous on $(0, \infty)$ and that $f(x) \geq 0$ on $[1, \infty)$.

Note that $f'(x) < 0$ on $[e, \infty)$

We choose $k = 3$

$\lim_{t\to\infty} \int_3^t \frac{\ln x}{x}\,dx = \lim_{t\to\infty} \frac{1}{2}\ln^2(x)\Big|_3^t = \infty.$

Therefore the series diverges by Integral Test

🐳 **Info — p-Series Test**

If $p > 1$, then the p —series $\sum_{n=0}^\infty \frac{1}{n^p}$ converges

If $p \leq 1$, then the p —series $\sum_{n=0}^\infty \frac{1}{n^p}$ diverges

🐳 **Info — Direct Comparison Test**

Let k be a positive integer. Suppose $\{a_n\}_{n=1}^\infty, \{b_n\}_{n=1}^\infty$ be two sequences satisfying

$0 \leq a_n \leq b_n$, for ever $n \geq k$

If $\sum_{n=1}^\infty b_n$ converges, $\sum_{n=1}^\infty a_n$ converges

If $\sum_{n=1}^\infty b_n$ diverges, $\sum_{n=1}^\infty a_n$ diverges

Example:

1. $\sum_{n=1}^\infty \frac{n}{n^3+1}$

   Choose $b_n = \frac{1}{n^2}$

   Since $\sum_{n=1}^\infty \frac{1}{n^2}$ converge by p-series test, therefore $\sum_{n=1}^\infty \frac{n}{n^3+1}$ converges by DCT
2. $\sum_{n=0}^\infty \frac{3^n+1}{2^{n+1}-1}$

   Choose $b_n = \frac{1}{2}\left(\frac{3}{2}\right)^n < \frac{3^n+1}{2^{n+1}-1}$

   SInce $\sum_{n=0}^\infty b_n = \frac{1}{2}\left(\frac{3}{2}\right)^n$ diverges by GST with $|\frac{3}{2}| > 1$, therefore $\sum_{n=0}^\infty \frac{3^n+1}{2^{n+1}-1}$ diverges by DCT

Info — Limit Comparison Test

Let $\{a_n\}_{n=1}^{\infty}, \{b_n\}_{n=1}^{\infty}$ be sequences and suppose that there is a positive integer k s.t.

$a_n \geq 0$ and $b_n \geq 0, \forall n \geq k$

We obtain $\lim_{n\to\infty} \frac{a_n}{b_n} = L$

1. $0 < L < \infty$ then $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ both converges or both diverges
2. $L = 0$, then
   - if $\sum_{n=1}^{\infty} b_n$ converges, $\sum_{n=1}^{\infty} a_n$ converges
   - if $\sum_{n=1}^{\infty} a_n$ diverges, $\sum_{n=1}^{\infty} b_n$ diverges
3. $L = \infty$, then
   - if $\sum_{n=1}^{\infty} a_n$ converges, $\sum_{n=1}^{\infty} b_n$ converges
   - if $\sum_{n=1}^{\infty} b_n$ diverges, $\sum_{n=1}^{\infty} a_n$ diverges

Note that if the limit is DNE and not ∞, it is inconclusive (i.e. oscillating)

Info — Alternating Series Test

Let $\{a_n\}_{n=1}^{\infty}$ be a sequence. If

1. $\lim_{n\to\infty} a_n = 0$
2. and there is a positive integer k s.t. $a_n \geq a_{n+1} > 0, \forall n \geq k$

then the series $\sum_{n=1}^{\infty} (-1)^{n-1} a_n$ and $\sum_{n=1}^{\infty} (-1)^n a_n$ both converge.

We prove this for $\sum_{n=1}^{\infty} (-1)^{n-1} n$ and $k = 1$.

Suppose that

- $a_n > -, \forall n \geq 1$
- $a_n \geq a_{n+1}, \forall n \geq 1$
- $\lim_{n\to\infty} a_n = 0$

For $k \geq 2$:

$S_{2k} = S_{2k-2} + a_{2k-1} - a_2 k \geq S_{2k-2}$

$S_{2k} = a_1 - a_2 + a_3 - a_4 + \ldots - a_{2k-2} + a_{2k-1} - a_{2k}$

$= a_1 - (a_2 - a_3) - (a_4 - a_5) - \ldots - (a_{2k-2} - a_{2k-1}) - a_{2k} \leq a_1$

So we have $S_2 \leq S_4 \leq \ldots \leq S_{2k} \leq \ldots \leq a$

By MCT, this sequence converges

For $k \geq 1$:

$S_{2k+1} = S_{2k} + a_{2k+1}$

$\lim_{k\to\infty} S_{2k+1} = L$

By MCT, this sequence converges

Info — Alternating Series Estimation

Let $\{a_n\}_{n=1}^{\infty}$ be a sequence. If

1. $\lim_{n\to\infty} a_n = 0$
2. and there is a positive integer k s.t. $a_n \geq a_{n+1} > 0, \forall n \geq k$

Let $S_m = \sum_{n=1}^{\infty} (-1)^{n-1} a_n$ and $T = \sum_{n=1}^{\infty} (-1)^n a_n$

Then for all integers $m \geq k$, we have the following:

4. $|S - S_m| \leq a_{m+1}$
5. If m is even, then S_m is an underestimate of S and T_m is an overestimate of T
6. If m is odd, then S_m is an overestimate of S and T_m is an underestimate of T

Example:

$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$

Is proven to be convergent by AST

For $m = 5 : |S - S_5| \leq a_6 = \frac{1}{6}$. It is an overestimation.

To guarantee an error be 0.001, $\frac{1}{m+1} \leq \frac{1}{1000} \implies m \geq 999$

Info — Absolute Convergence Test

If the series $\sum_{n=1}^{\infty} |a_n|$ converges, then the series $\sum_{n=1}^{\infty} a_n$ converges.

Note that the converse is false.

A series is **absolutely convergent** if the $\sum_{n=1}^{\infty} |a_n|$ converges.

A series is **conditionally convergent** if it converges but $\sum_{n=1}^{\infty} |a_n|$ diverges

Tip — Series Classification

1. Try Divergence Test
2. Test $\sum |a_n|$
3. Test $\sum a_n$

Examples:

1. $\sum_{n=0}^{\infty} \frac{\cos(n)}{n^2}$ is absolutely convergent
2. $\sum_{n=1}^{\infty} (-1)^n \arctan\left(\frac{n}{n+1}\right)$

   $\lim_{n\to\infty} (-1)^n \arctan\left(\frac{n}{n+1}\right) = \text{DNE(oscillating)}$

   The series diverges by Divergence Test

3. $\sum_{n=1}^{\infty} \left(-\frac{2}{3}\right)^n$

   Divergence Test fails

$\sum_{n=1}^{\infty} |(-\frac{2}{3})^n| = \sum_{n=1}^{\infty} (\frac{2}{3})^n$

The series is geometric, converges by GST and is absolutely convergent.

4. $\sum_{n=1}^{\infty} \frac{(-1)^n \sqrt{n^2+n}}{n^{\frac{3}{2}}}$

   Divergence Test fails

   $\sum_{n=1}^{\infty} |\frac{(-1)^n \sqrt{n^2+n}}{n^{\frac{3}{2}}}| \implies \lim_{n\to\infty} \frac{\frac{\sqrt{n^2+n}}{n^{\frac{3}{2}}}}{\frac{1}{\sqrt{n}}} = 1$

   diverges by LCT.

   $\sum_{n=1}^{\infty} \frac{(-1)^n \sqrt{n^2+n}}{n^{\frac{3}{2}}}$ converves by AST

> **Info — Ratio Test**
> Let k be a positive integer. Suppose that $\{a_n\}_{n=1}^{\infty}$ is a sequence satisfying $a_n \neq 0$ for every $n \geq k$. Let
>
> $$L = \lim_{n\to\infty} |\frac{a_{n+1}}{a_n}|$$
>
> Further, supose that either $L \in \mathbb{R}$ or $L = \infty$
>
> 1. If $L < 1$, then $\sum_{n=1}^{\infty} a_n$ converges absolutely.
> 2. If $L > 1$ or if $L = \infty$, then $\sum_{n=1}^{\infty} a_n$ diverges.
> 3. If $L = 1$, then $\sum_{n=1} \infty a_n$ is **inconclusive**

Examples:

1. $\sum_{n=1}^{\infty} \frac{(-1)^n 2^n}{n^2 3^n}$

   $\lim_{n\to\infty} |\frac{(-1)^{n+1} 2^{n+1}}{(n+1)^2 3^{n+1}} \cdot \frac{n^2 3^n}{(-1)^n 2^n}| = \lim_{n\to\infty} \frac{2n^2}{3(n+1)^2} = \frac{2}{3} < 1$

   Converges absolutely by Ratio Test

2. $\sum_{n=1}^{\infty} \frac{b^n}{n!}$

   $\lim_{n\to\infty} |\frac{b^{n+1}}{(n+1)!} \cdot \frac{n!}{b^n}| = \lim_{n\to\infty} \frac{b}{n+1} = 0 < 1$

   Converges absolutely by Ratio Test

3. $\sum_{n=1}^{\infty} \frac{n^n}{n!}$

   $\lim_{n\to\infty} |\frac{(n+1)^{n+1}}{(n+1)!} \cdot \frac{n!}{n^n}| = \lim_{n\to\infty} (\frac{n+1}{n})^n = 1$

   Inconclusive by Ratio Test

> **Info — n-th Root Test**
> Let $L = \lim_{n\to\infty} \sqrt[n]{|a_n|}$ and suppose that $L \in \mathbb{R}$ or $L = \infty$
>
> 1. If $L < 1$, then $\sum_{n=1}^{\infty} a_n$ converges absolutely.
> 2. If $L > 1$ or if $L = \infty$, then $\sum_{n=1}^{\infty} a_n$ diverges.
> 3. If $L = 1$, then $\sum_{n=1} \infty a_n$ is **inconclusive**

Summary

Summary of Chapter 3

- Partial sums: $S_m = \sum_{n=1}^{m} a_n$ and $\sum_{n=1}^{\infty} a_n = \lim_{m\to\infty} S_m$ if this limit exists.
- Calculating sums of geometric series and telescoping series.
- Convergence tests:
  - Divergence test (Check $\lim_{n\to\infty} a_n$)
  - Geometric series test and p-series test
    (Convergence/divergence is known for all such series.)
  - Comparison test and limit comparison test
    (Can you compare $\sum |a_n|$ to a geometric series or a p-series?)
  - Ratio test and root test
    (May be helpful if the terms involve factorials, exponentials, powers)
  - Integral test
    (Can we apply this test to the series $\sum |a_n|$?)
  - Absolute convergence test
    (If $\sum |a_n|$ converges, then $\sum a_n$ converges.)
  - Alternating series test
    (If $\lim_{n\to\infty} a_n = 0$ and $\sum |a_n|$ diverges, then we hope this test applies!)
- Absolute versus conditional convergence.